

Wenyu Chiou

AI Agent Infrastructure & Evaluation Engineer

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I build governed, fail-closed LLM-agent infrastructure — anti-fabrication gates, delegation contracts, and audit trails — and the empirical simulations that stress-test whether AI agents can stand in for real decision-makers.

— SELECTED SYSTEMS

research-hub MAINTAINED OPEN-SOURCE TOOL

Local-first research-literature pipeline (Zotero / Obsidian / NotebookLM) whose write path is guarded by a fail-closed anti-fabrication gate: references that cannot be verified are quarantined with an explicit failure taxonomy, never silently accepted. Sole designer and maintainer end to end — architecture, authenticity gate, MCP server, CI design, evaluation suite, release process.

PyPI research-hub-pipeline v1.1.1 · MCP server · CI across 3 OS × 4 Python versions · in-repo retrieval-recall evaluation suite against golden fixtures

fable-method-harness MAINTAINED OPEN-SOURCE TOOL

A portable operating harness for AI-assisted work: model/cost routing, honest-failure discipline, and an audit trail for every step. Measured against a benchmark suite rather than asserted; verified across Claude and Codex runtimes with reserved slots for others.

validation CI · benchmark + distillation harness · routing playbook · audit-trail memory

agent-collab-skills MAINTAINED OPEN-SOURCE TOOL

A multi-agent orchestration toolkit: task splitter, output reconciler, adversarial debate, shared memory, and a pre-merge acceptance gate. Composes with the delegation wrappers below to run bounded multi-agent rounds without silent drift.

test suite + CI · catalog & context-policy tests · composes with codex-delegate

codex-delegate MAINTAINED OPEN-SOURCE TOOL

Cross-platform agent-delegation wrappers with anti-fabrication contracts — change attribution is verified from git state, not the delegate's own report. A regression test pins a real upstream stdin hang and fails without the fix.

CI on Ubuntu and Windows across multiple Python versions · contract + failure-mode tests

WAGF — Water Agent Governance Framework IN PREPARATION FOR RELEASE

A governance layer for LLM-driven agents: agent decisions are validated against physical constraints and behavioral theory, with retry feedback and audit logs, before they update simulation state — fail-closed, no silent passes.

domain-rule + behavioral-theory validation · retry feedback · audit logs

— OPEN-SOURCE CONTRIBUTIONS

- 10 merged pull requests in third-party open-source projects, including a root-caused engine-selection bug fix with a regression test that fails on main (BuilderIO/agent-native ×4) and a Windows file-permissions security fix (langchain-ai/openwiki).
- Taiwan-market data features across three merged PRs (ZhuLinsen/daily_stock_analysis ×3, re-scoped through maintainer review); pure-bash UTF-8 / East-Asian-width display fix (Nanako0129/coralline); documentation fix (NVIDIA/skills).

— PUBLICATION

Yang, Y. C. E., & Chiou, W. (2026). Leveraging Large Language Models for Agent-Based Simulation of Human–Water System Interactions. *Water Resources Research*, 62(6), e2025WR042111. doi.org/10.1029/2025WR042111 Published · second author of two.

— DOMAIN & RISK ANALYTICS

FLOODABM — 52,141-household coupled agent-based × catastrophe flood model with NFIP premium, payout, and deductible mechanics; calibrated on a self-collected 937-household survey (Passaic River Basin, NJ) via structural equation modeling and Bayesian calibration; simulated payouts validated against OpenFEMA insurance-claims data. Research prototype, archived.

Catastrophe-risk modeling fluency: NFIP mechanics, exceedance-probability curves, coupled ABM × catastrophe modeling.

— COMMUNITY

awesome-agentic-ai-zh — trilingual agentic-AI learning curriculum with CI-enforced locale parity and external contributors; 4.5k+ GitHub stars (July 2026).

— EDUCATION

Ph.D., Civil & Environmental Engineering , Lehigh University	2024 – present · expected 2028
M.S., Hydrological and Oceanic Sciences , National Central University	Sep 2021 – Jun 2023
B.S., Earth Sciences , National Central University	Sep 2017 – Jun 2021

— SKILLS

AI-agent systems: LLM-agent orchestration · multi-agent collaboration · evaluation design · anti-fabrication & governance · MCP / agent tooling · local model deployment (Ollama)

Programming & tooling: Python · testing & CI · Git · cross-platform (Windows / POSIX) engineering

Statistical modeling: structural equation modeling (SEM) · Bayesian calibration

Risk analytics: NFIP mechanics · exceedance-probability curves · coupled catastrophe modeling